Cache miss types

Sparsh Mittal

Courtesy for some slides: Dr. Rajeev Balasubramonian

How much associativity?

Increased associativity decreases miss rate

But with diminishing returns

Example miss-rate results on a 64KB D-cache

1-way: 10.3%
2-way: 8.6%
4-way: 8.3%
8-way: 8.1%

➔ With increasing associativity, miss-rate reduces but reach a plateau

➔ It is possible that an access misses in fully-associative cache but hits in direct-mapped cache or set-associative cache.

Types of Misses

Compulsory Misses

Misses that happen when we read in a piece of data for the first time.

Conflict Misses

Misses that occur due to the limited amount of associativity in a set associative or direct mapped cache. A fully associative (FA) cache has no conflict misses.

Example: Assume that 5 blocks (accessed by the program) map to the same set in a 4-way associative cache. Only 4 out of 5 can be kept.

Capacity Misses

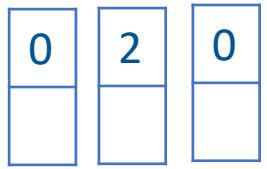
Misses that occur due to the limited size of a cache. Example: Assume the working set of a program is 10 KB, and the cache size is 8 KB.

Example to understand conflict misses

Assume a direct-mapped cache with 2 sets. Block-size 1B

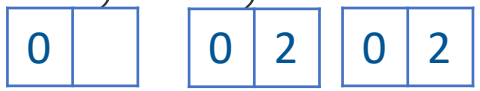
The accesses sequence is 0, 2, 0

Result:



All accesses miss.

If we had a 2-way FA cache with 1 set. 0 , 2, 0



Last access hits!

➔ In direct-mapped cache, miss on 2nd '0' access was conflict miss.

Example to understand capacity miss

Assume a direct-mapped cache with 2 sets. Block-size 1B

The accesses sequence is 0, 2, 4, 0

Result:

0	2	4	0

All accesses miss.

If we had a 2-way FA cache with 1 set. 0 , 2, 4, 0

0		0	2	4	2	4	0

All accesses miss!

➔ In direct-mapped cache, miss on 2nd '0' access was capacity miss. This miss happened because of limited capacity; not because of limited associativity.

Algorithm for distinguishing 3 types of misses

For each address: Access the given cache. If it is a hit, fine. Else, it is a miss.

-- If the address is coming for the first time, the miss is a compulsory miss.

-- Else, the miss is either a capacity miss or conflict miss.

Now, take the fully-associative cache of same block size and same total-size.

For each address: Access this cache. If it is a hit, fine. Otherwise, it is a miss.

-- If the address is coming for the first time, the miss is a compulsory miss.

-- Else, it is a capacity miss (fully-associative cache has NO conflict miss).

If an access missed in both given cache and FA cache, it was a capacity miss.

If an access missed in given cache, but not in FA cache, it was a conflict miss.

Example 1

For understanding difference between conflict and capacity misses

Solved example

Consider a direct mapped cache of size 4 blocks with block size 4 Bytes. If the following memory addresses are requested, then how many compulsory, capacity and conflict miss are there respectively ?
0, 4, 256, 4, 268, 4, 8, 0, 256, 4

Address sub-division:
[Tag bits | SetID | Block Offset]
Let us find the number of bits in each of these parts.
Block Size = 4 Bytes = 2^2 Bytes, So Block Offset = **2 bits**

No. of Blocks = 4 = 2^2, So for SetID we would use **2 bits**

Address division with direct-mapped cache

Request	Binary	Tag	SetID	Offset
0	0000 0000	0000	00	00
4	0000 0100	0000	01	00
8	0000 1000	0000	10	00
256	1 0000 0000	10000	00	00
268	1 0000 1100	10000	11	00

Cache state (1 of 4)

After 0

0

Compulsory miss

After 4

0
4

Compulsory miss

After 256

256
4

Compulsory miss

Cache state (2 of 4)

After 4

256
4

Hit

After 268

256
4
268

Compulsory miss

After 4

256
4
268

Hit

Cache state (3 of 4)

After 8

256
4
8
268

Compulsory Miss

After 0

0
4
8
268

? Miss

After 256

256
4
8
268

? Miss

- Misses marked with ? can be either capacity or conflict misses.
- They cannot be compulsory because 0 and 256 have already come before.
- To distinguish which type it is, we have to test the address sequence with a fully-associative cache also.

Cache state (4 of 4)

After 4

256
4
8
268

Hit

Results with fully-associative cache

We now consider a fully associative cache of size 4 blocks with block size 4 Bytes. We will feed the same access sequence as before
Address sub-division for fully-associative cache is as follows
[Tag bits | Block Offset]

Address	Binary	Tag	Offset
0	0000 0000	000000	00
4	0000 0100	000001	00
8	0000 1000	000010	00
256	1 0000 0000	1000000	00
268	1 0000 1100	1000011	00

Test with fully-associative cache

0, 4, 256, 4, 268, 4, 8, 0, 256, 4

After 0	0				Miss
After 4	4	0			Miss
After 256	256	4	0		Miss
After 4	4	256	0		Hit
After 268	268	4	256	0	Miss
After 4	4	268	256	0	Hit
After 8	8	4	268	256	Miss
After 0	0	8	4	268	Miss
After 256	256	0	8	4	Miss
After 4	4	256	0	8	Hit

They missed with fully-associative cache also, hence, they cannot be conflict misses.

Conclusion from fully-associative cache

A fully-associative cache has no conflict miss. It has only compulsory and capacity misses.

Since 0 and 256 are missing in fully-associative cache and they cannot be compulsory, the only possibility is that they are capacity misses.

Lets discuss second example

Example 2

Consider a direct mapped cache of size 4 blocks with block size 4B
The access sequence is: 0, 4, 256, 4, 4, 8, 0, 256, 4
Find whether each miss is compulsory or conflict or capacity.

With direct-mapped cache

After 0

0

Compulsory miss

After 4

0
4

Compulsory miss

After 256

256
4

Compulsory miss

With direct-mapped cache

After 4

256
4

Hit

After 4

256
4

Hit

After 8

256
4
8

Compulsory miss

With direct-mapped cache

After 0

0
4
8

?Miss

Conflict miss

By our analysis with fully-associative cache (shown on next slide), we know that they do not miss in FA-cache, Hence, both these misses are conflict misses.

After 256

256
4
8

?Miss

Conflict miss

After 4

256
4
8

Hit

Example 2

0, 4, 256, 4, 4, 8, 0, 256, 4

Fully-associative:

0	miss 0 : compulsory
4 0	miss 4 : compulsory
256 4 0	miss 256 : compulsory
4 256 0	hit 4
4 256 0	hit 4
8 4 256 0	miss 8 : compulsory
0 8 4 256	hit 0
256 0 8 4	hit 256
4 256 0 8	hit 4

Getting more insights

- Lets assume you have a 32KB direct mapped cache, and consider the following 2 cases:

1. You repeatedly iterate over a 128KB array. There's no way the data can fit in that cache, therefore all the misses are capacity ones
2. You have 2 small 8KB arrays, but they are both aligned and map to the same sets => while they could theoretically fit in the cache (if you make them map to different sets), they will not utilize the full cache size and instead compete for the same group of sets and thrash each other. These are conflict misses, since the data could fit, but still collides due to organization.

Insights into compulsory miss

- Consider a cache which has 1 set and 2 ways. Block size is 4B. It gets accesses: 0, 4, 8, 1
- Now, access to 0 will bring Block [0,1,2,3] in first way of cache.
- Access to 4 will bring Block [4,5,6,7] in cache.
- Access to 8 will bring Block [8,9,10,11] in cache and will replace block [0,1,2,3].
- Now, when 1 is accessed, it will lead to miss. An important point: it is not considered compulsory miss, because block with address 1 was already fetched earlier in cache.

Cache Design Trade-offs

Design change	Effect on miss rate	Negative performance effect
Increase cache size	Decrease capacity misses	May increase access time
Increase associativity	Decrease conflict misses	May increase access time
Increase block size	Decrease compulsory misses	Increases miss penalty. For very large block size, may increase miss rate due to pollution.

How higher block-size reduces compulsory misses

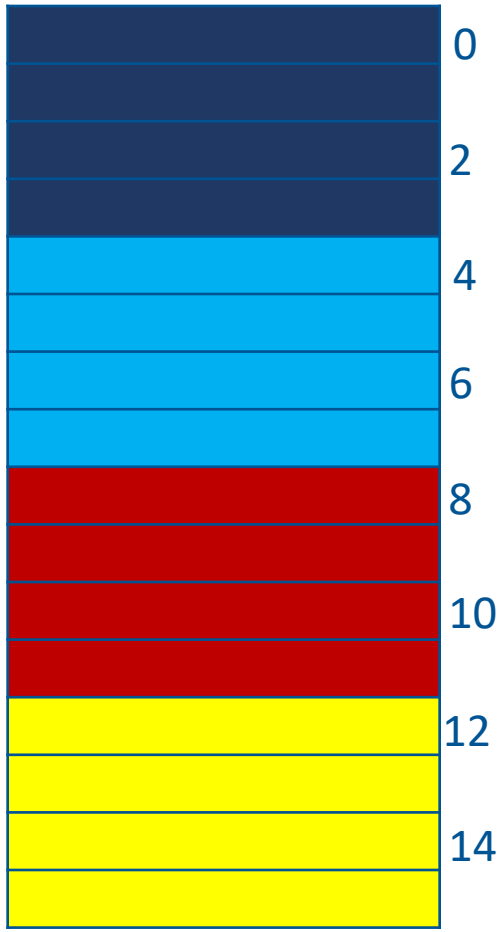
- Block size is 4B (block-offset has 2 bits)

- If I access address 7, which addresses I get?
- Answer: 4, 5, 6 and 7

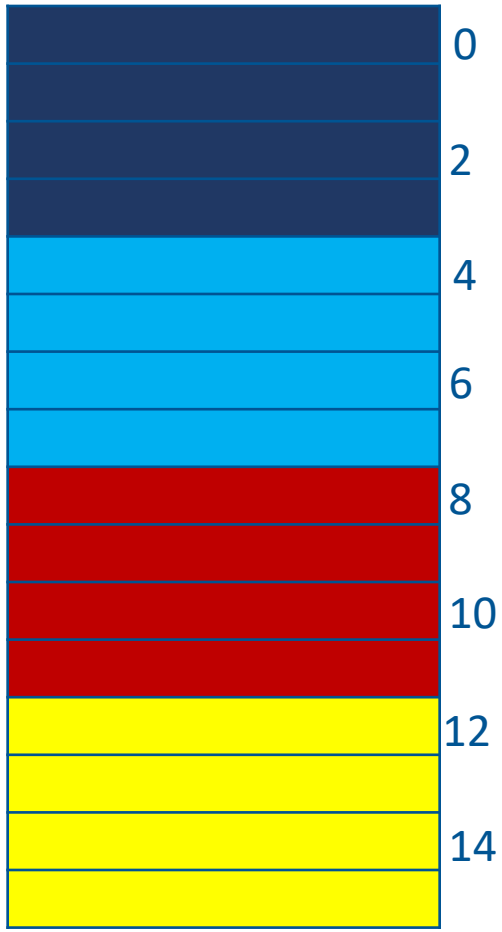
4=**01**<u>00</u>, 5=**01**<u>01</u> 6=**01**<u>10</u>, 7=**01**<u>11</u>

On ignoring offset bits, remaining bits are same.

4, 5, 6 and 7 are part of the same block



How higher block-size reduces compulsory misses

- If I first access address 7 and then 4, will access to 4 lead to a hit?
- Answer: YES.
- In this way, use of bigger cache block-size has avoided compulsory miss to ‘4’.

- If we had used block-size of 1B, then, miss to ‘4’ could not be avoided.